12. DEMONSTRATE CYBER EVASION TECHNIQUES

DATE: 08/04/2026

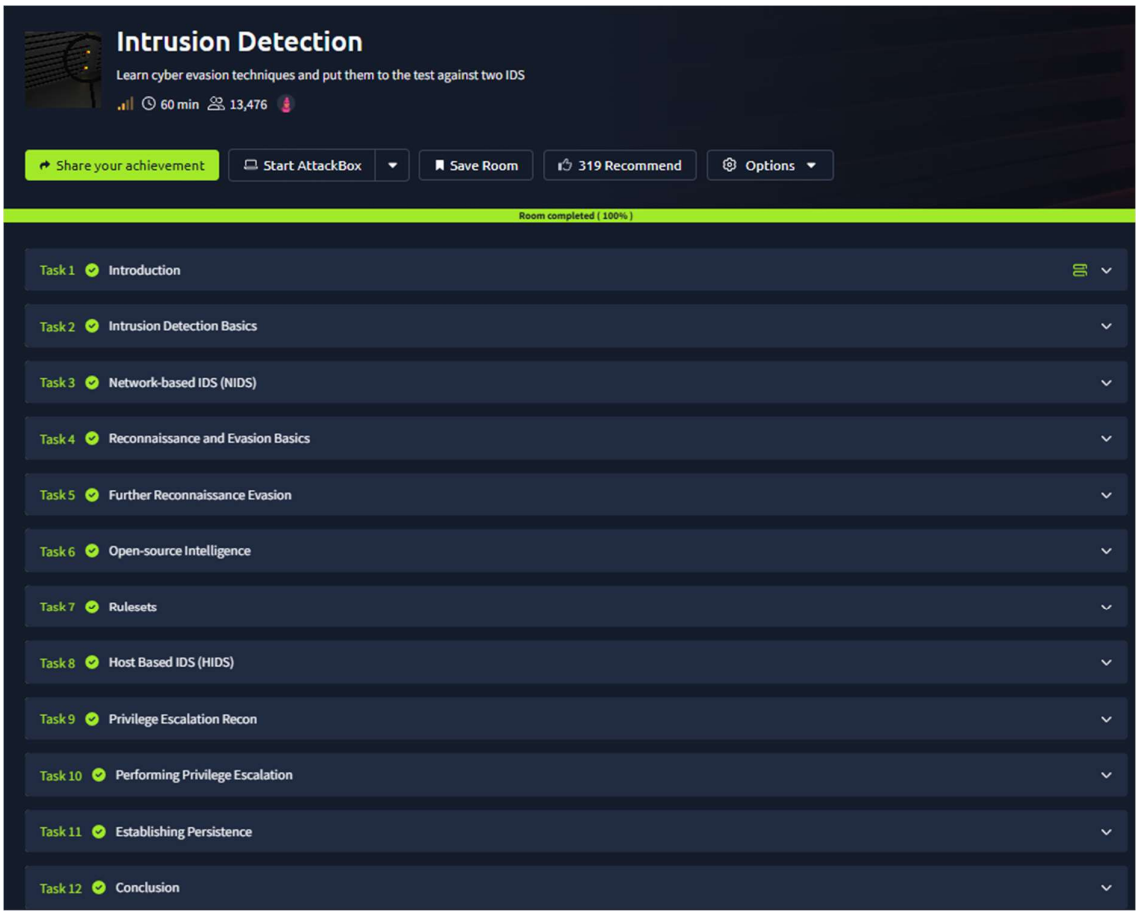
Aim:

To learn cyber evasion techniques and put them to the test against two IDS.

Procedure:

1. Log into TryHackMe and open the Intrusion Detection room.
2. Study the basics of Intrusion Detection Systems (IDS) and their role in security.
3. Understand common detection methods such as signature-based and anomaly-based detection.
4. Learn cyber evasion techniques used to bypass IDS.
5. Analyze how different IDS respond to various attack patterns.
6. Perform tasks to test evasion techniques against the IDS.
7. Review results and confirm completion.

Tasks:

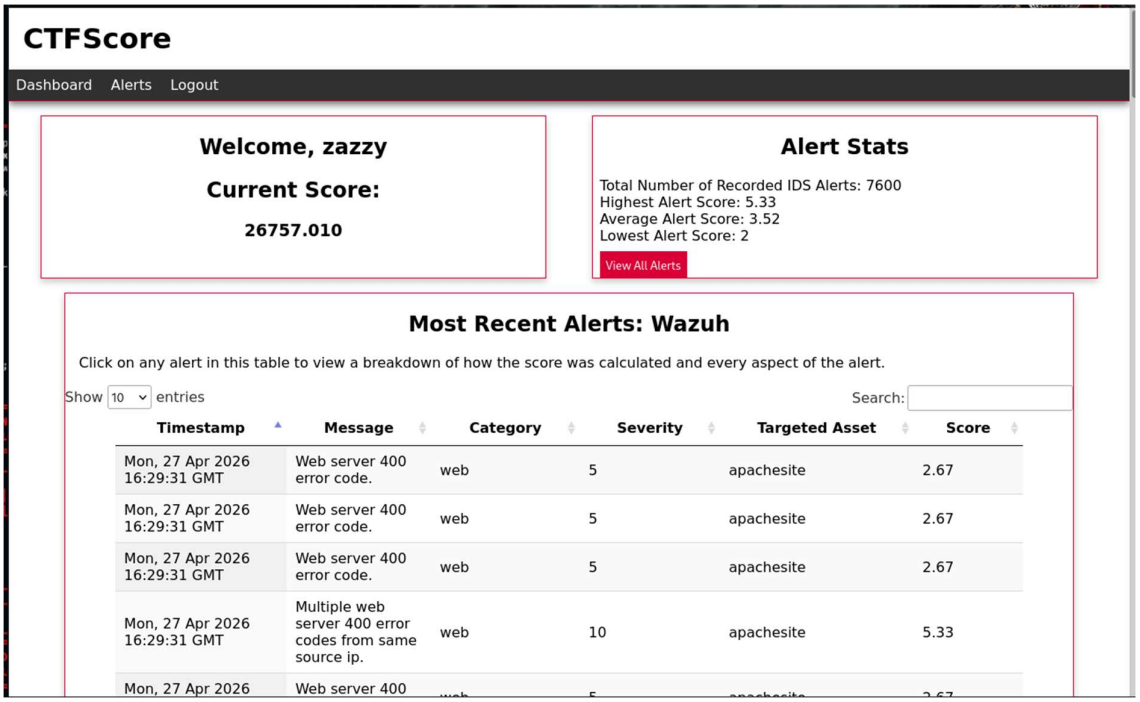


TASK 1 - Introduction

Answer the questions below

Deploy the target machine and create an account and log into the system at http://MACHINE_IP:8000, in preparation for future tasks.

No answer needed ✓ Correct Answer



Timestamp	Message	Category	Severity	Targeted Asset	Score
Mon, 27 Apr 2026 16:29:31 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:31 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:31 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:31 GMT	Multiple web server 400 error codes from same source ip.	web	10	apachesite	5.33
Mon, 27 Apr 2026	Web server 400				

TASK 2 – Intrusion Detection Basics

Answer the questions below

What IDS detection methodology relies on rule sets?

signature-based detection ✓ Correct Answer

TASK 3 -Network based IDS (NIDS)

Answer the questions below

What widely implemented protocol has an adverse effect on the reliability of NIDS?

TLS ✓ Correct Answer

Experiment by running tools against the target and viewing the resultant alerts. Is there any unexpected activity?

No answer needed ✓ Correct Answer

TASK 4 – Reconnaissance and Evasion Basics

Answer the questions below

What scale is used to measure alert severity in Suricata? (*-*)

1-3 ✓ Correct Answer

How many services is nmap able to fully recognise when the service scan (-sV) is performed?

3 ✓ Correct Answer

TASK 5 – Further Reconnaissance Evasion

Answer the questions below

Nikto, should find an interesting path when the first scan is performed, what is it called?

/login ✓ Correct Answer

What value is used to toggle denial of service vectors when using scan tuning (-T) in nikto?

6 ✓ Correct Answer

Which flags are used to modify the request spacing in nikto? Use commas to separate the flags in your answer.

6,A,B ✓ Correct Answer

TASK 6 – Open Source Intelligence

Answer the questions below

What version of Grafana is the server running?

8.2.5 ✓ Correct Answer

What is the ID of the severe CVE that affects this version of Grafana?

CVE-2021-43798 ✓ Correct Answer

If this server was publicly available, What site might have information on its services already?

shodan ✓ Correct Answer

How would we search the site "example.com" for pdf files, using advanced Google search tags?

site:example.com filetype:pdf ✓ Correct Answer

TASK 7 - Rulesets

```
#################################### Security ####################################
[security]
# disable creation of admin user on first start of grafana
;disable_initial_admin_creation = false

# default admin user, created on startup
admin_user = grafana-admin

# default admin password, can be changed before first start of grafana,  or in prof
ile settings
admin_password = GraphingTheWorld32

# used for signing
;secret_key = SW2YcwTIb9zpOOhoPsMm

# disable gravatar profile images
;disable_gravatar = false
```

Most Recent Alerts: Suricata

Click on any alert in this table to view a breakdown of how the score was calculated and every aspect of the alert.

Show 10 entries Search:

Timestamp	Message	Category	Severity	Targeted Asset	Score
Mon, 27 Apr 2026 16:31:33 GMT	SURICATA STREAM Packet with invalid timestamp	Unknown Classtype	3	172.200.0.30	5.33
Mon, 27 Apr 2026 16:31:33 GMT	SURICATA STREAM Packet with invalid timestamp	Unknown Classtype	3	172.200.0.30	5.33
Mon, 27 Apr 2026 16:31:33 GMT	SURICATA STREAM Packet with invalid timestamp	Unknown Classtype	3	172.200.0.30	5.33
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Mon, 27 Apr 2026 16:31:33 GMT	SURICATA STREAM Packet with invalid timestamp	Unknown Classtype	3	172.200.0.30	5.33
Mon, 27 Apr 2026 16:31:33 GMT	SURICATA STREAM Packet with invalid timestamp	Unknown Classtype	3	172.200.0.30	5.33

Answer the questions below

What is the password of the grafana-admin account?

GraphingTheWorld32 ✓ Correct Answer

Is it possible to gain direct access to the server now that the grafana-admin password is known? (yay/nay)

yay ✓ Correct Answer

Are any of the attached IDS able to detect the attack if the file /etc/shadow is requested via the exploit, if so what IDS detected it?

Suricata ✓ Correct Answer

TASK 7 – Host Based IDS (HIDS)

Most Recent Alerts: Wazuh

Click on any alert in this table to view a breakdown of how the score was calculated and every aspect of the alert.

Show 10 entries Search:

Timestamp	Message	Category	Severity	Targeted Asset	Score
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67
Mon, 27 Apr 2026 16:29:55 GMT	Web server 400 error code.	web	5	apachesite	2.67

Answer the questions below

What category does Wazuh place HTTP 400 error codes in?

web ✓ Correct Answer

Play around with some post-exploitation tools and commands and make note of what activity is detected by Wazuh; compare it to the activity that's detected by Suricata.

No answer needed ✓ Correct Answer

TASK 9 – Privilege Escalation Recon

Most Recent Alerts: Wazuh

Click on any alert in this table to view a breakdown of how the score was calculated and every aspect of the alert.

Show 10 entries Search:

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Mon, 27 Apr 2026 16:29:53 GMT	Web server 400 error code.	web	5	apachesite	2.67

Answer the questions below

What tool does linPEAS detect as having a potential escalation vector?

docker ✓ Correct Answer

Is an alert triggered by Wazuh when linPEAS is added to the system, if so what its severity?

5 ✓ Correct Answer

TASK 10 – Performing Privilege Escalation

Answer the questions below

Perform the privilege escalation and grab the flag in /root/

{SNEAK_ATTACK_CRITICAL} ✓ Correct Answer

TASK 11 – Establishing Persistence

Answer the questions below

Abuse docker to establish a backdoor on the host system

No answer needed ✓ Correct Answer

TASK 12 - Conclusion

Result:

Thus the TryHackMe room “Intrusion Detection” was completed successfully.